

24.

$$a) \quad C+S = \begin{pmatrix} 2 & 8 & 4 \\ 1 & 11 & 2 \\ 3 & 9 & 1 \\ 2 & 7 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 4 & 1 \\ 1 & 6 & 2 \\ 0 & 7 & 1 \\ 2 & 9 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 12 & 5 \\ 2 & 17 & 4 \\ 3 & 16 & 2 \\ 4 & 16 & 5 \end{pmatrix}$$

C+S is the total sales for both stores (city and suburban) of each musical instruments week by week for January 2014

$$b) \quad 100C = \begin{pmatrix} 200 & 800 & 400 \\ 100 & 1100 & 200 \\ 300 & 900 & 100 \\ 200 & 700 & 300 \end{pmatrix}$$

$$c) \quad P = \begin{pmatrix} 200 & 800 & 400 \\ 100 & 1100 & 200 \\ 300 & 900 & 100 \\ 200 & 700 & 300 \end{pmatrix} + \begin{pmatrix} 90 & 360 & 90 \\ 90 & 540 & 180 \\ 0 & 630 & 90 \\ 180 & 810 & 180 \end{pmatrix}$$

$$= \begin{pmatrix} 290 & 1160 & 490 \\ 190 & 1640 & 380 \\ 300 & 1530 & 190 \\ 380 & 1510 & 480 \end{pmatrix}$$

d) P is the week by week total profit made by both stores for each instrument in Jan 2014

$$e) \quad \$8540$$

$$f) \quad D = \begin{pmatrix} 200 & 800 & 400 \\ 100 & 1100 & 200 \\ 300 & 900 & 100 \\ 300 & 700 & 300 \end{pmatrix} - \begin{pmatrix} 90 & 360 & 90 \\ 90 & 540 & 180 \\ 0 & 630 & 40 \\ 180 & 810 & 180 \end{pmatrix}$$

$$= \begin{pmatrix} 110 & 440 & 310 \\ 10 & 560 & 20 \\ 300 & 270 & 10 \\ 20 & -110 & 120 \end{pmatrix}$$

D is the difference in profit made by the city stores (over the suburban store) for each musical instrument over the 4 weeks in Jan 2014

- g) Suburban store's profit on keyboard in week 4 exceeded the city store's profit by \$110.

28. a)

		To			
		A	B	C	D
From	A	0	1	0	3
	B	0	0	1	0
	C	0	0	0	1
	D	3	0	0	0

b) 9

$$c) P^2 = \begin{pmatrix} 9 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 9 \end{pmatrix}$$

d) Number of 2-steps (2-steps) routes that exist between the four locations

e) There are 3 2-stage routes from C to A.

$$f) P^3 = \begin{pmatrix} 0 & 9 & 0 & 28 \\ 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 9 \\ 27 & 0 & 3 & 0 \end{pmatrix}$$

No. of 3-steps (3-steps) routes that exist between the four locations.